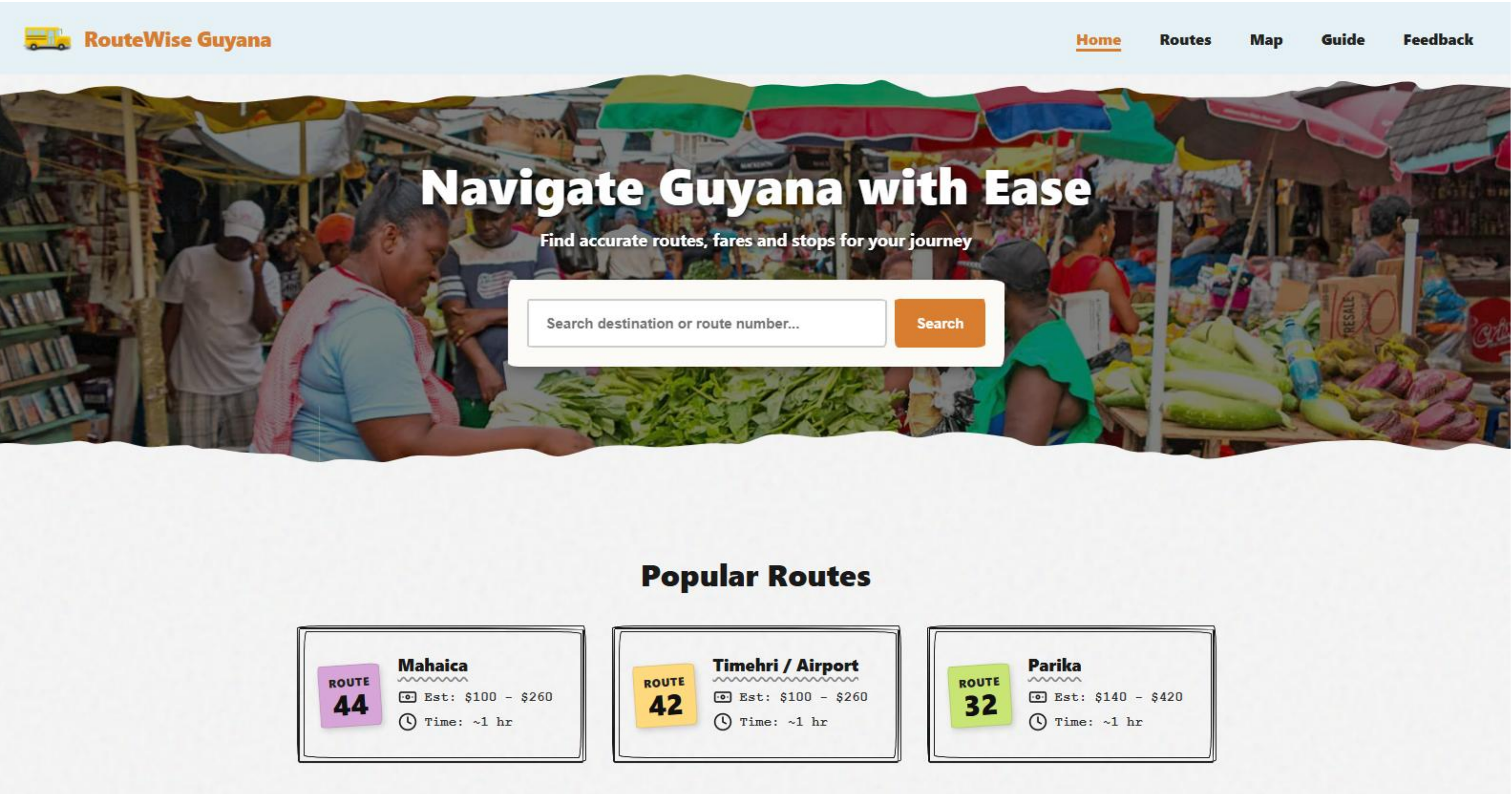
FACULTY OF NATURAL SCIENCES
DEPARTMENT OF COMPUTER SCIENCE

CSE 2201: Internet Computing I
Group Project: Web Design and Development
RouteWise Guyana: A Web-Based Minibus Route and Fare Information System



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1. Introduction

Public transportation in Guyana plays a significant role in the daily movement of people, particularly with privately operated minibuses. Minibuses are used by many people traveling around Georgetown and across surrounding commuter corridors such as East Coast Demerara, East Bank Demerara, West Coast Demerara, West Bank Demerara, the Linden Highway and Berbice.

Despite the importance of this system, there appears to be little to no structured digital platform that provides reliable and accurate information about minibus routes, fare costs or pickup locations. Most commuters depend on informal methods such as asking the bus driver, family members, friends, conductors, or other passengers for guidance to determine how they should proceed.

Although informal methods may work for those familiar with the system in place, they causes considerable confusion among students, first time riders, and visitors unfamiliar with the minibus system. Because of this limited access to information, commuters may experience delays, select routes incorrectly, and be uncertain about fare costs. Additionally, in certain instances, commuters may be overcharged or take longer routes than required.

To address this issue, **RouteWise Guyana**, a web-based platform was designed to provide clear and accessible information on minibus routes, fares and key pickup points. This platform aims to improve the total commuting experience for all commuters by developing an intuitive user-friendly interface that allows all users to easily search for routes, review relevant details and interact with an integrated map.

It also demonstrates how client-side web technologies along with basic server side functionality solve real-world information problems with local context.

2. Problem Significance

Limited access to reliable public transportation information in Guyana goes beyond inconvenience; it affects how easily people move through their daily lives. For many commuters, minibuses provide the primary means of transporting themselves to school, work or errands, but unclear routes, fares, or stops often lead to confusion.

For example, someone unfamiliar with a route might board the wrong bus or arrive late simply because they were not sure where to go. These issues may seem small, but over time, they can reduce commuter confidence and slow people down.

From a usability perspective, the system may “work”, but inefficiently. The user may eventually arrive at their destinations, but often with extra effort or by asking others for help. Systems should support users in completing tasks with efficiency and satisfaction.

This becomes even more noticeable for new users, like tourists or people unfamiliar with the area, who may find the system difficult to navigate and therefore unable to utilize the system properly.

A platform like RouteWise Guyana could provide a solution to these problems by providing clear organized transportation information that allows users to plan routes and travel with greater confidence.

3. Requirement Analysis

To address the identified problem, a set of functional and non-functional requirements was established based on the needs of commuters in Guyana, the intended purpose of the system, and scope of the project. These requirements were used as guidelines for designing and implementing the solution and ensured that the system is user-friendly, practical and consistent with how it is going to be used in everyday life.

3.1 Functional Requirement

The system shall:

1. Allow users to search for minibus routes based on a specific starting point and destination.
2. Display the name of the route, the corridor, major stops, estimated fare and the expected travel time.
3. Provide an interactive map with route lines and markers to visually represent chosen routes.
4. Allow users to filter routes based on important corridors, including Georgetown, East Coast Demerara, East Bank Demerara, West Coast Demerara, West Bank Demerara, the Linden Highway, and Berbice.
5. Display commonly recognized landmarks or frequently visited places along routes to help users who might not be familiar with street names.
6. Allow users to select and view detailed information for specific routes.
7. Provide a structured list or directory of available routes for browsing purposes.
8. Allow users to submit feedback regarding incorrect, unclear or incomplete route information, including reports of overcharging.
9. Provide users with guidance on how to interpret routes, particularly for first time users of the transport system.
10. Allow users to access route information and fare information without requiring login or account creation.
11. Clearly indicate when route, fare, or travel-time information is estimated, incomplete or unverified.

3.2 Non-Functional Requirements

The system shall:

1. Provide a clear and accessible user interface that supports efficient navigation and task completion.
2. Be responsive and accessible across a range of devices, including desktop and mobile devices.
3. Ensure satisfactory performance, such as quick loading times and seamless map component interaction.
4. Maintain consistency in layout, navigation, and presentation of information across all pages.
5. Provide clear feedback to user activities, such as verifying successful form submission or route selection.
6. Present information in a structured and readable format to support users with varying levels of familiarity with public transportation.
7. Operate using publicly available or user-collected data, ensuring that all data usage complies with ethical and legal considerations.
8. Minimize the number of user actions required to access key route and fare information, with the intended target being no more than three actions from the main route search.
9. Be built with a modular, maintainable architecture to accommodate route data expansion and upgrades in the future.

4. System Design

4.1 System Overview

RouteWise Guyana is designed as a web-based public transportation information system that helps users access minibus route and fare information in a more organized way. The goal of RouteWise Guyana is to provide valuable information about how to get around Georgetown and its immediate commuter corridors (East Coast Demerara, East Bank Demerara, West Coast Demerara, West Bank Demerara, the Linden Highway, and Berbice) to users.

The system is available through a web browser and works on both desktop and mobile devices. This is important because it offers flexibility to users who may access the website in different situations. A student, for instance, could access from their laptop at home and plan their route prior to travelling. In contrast, a commuter or visitor may access the site from their smartphone at a bus park or at the roadside when they require quick information.

The primary function of the system is to allow users to search for a route by entering or selecting a starting point and destination. Once a route is selected, the system provides the relevant details about the route such as the route name, corridor, major stops, estimated fare, estimated travel time, landmarks or popular places, and map-based route information. The website is designed to help solve the problem of limited access to clear and reliable public transportation information.

The system includes both client-side and basic server-side functionality. On the client-side, HTML is used to structure the pages, CSS is used for layout and visual design, and JavaScript is used for interaction, searching, filtering, sorting, dynamic route display, and map-related behavior. The server-side employs PHP to handle submissions from the feedback form, and stores submitted reports.

A general overview of the overall system's workflow includes:

1. The user accesses the website on a desktop or mobile browser.
2. The user searches for, or selects a route.
3. The system displays matching route information.
4. The user views route details, estimated fare, travel time, landmarks, and map information.
5. The user may submit feedback if the information appears incorrect, unclear or incomplete data.

4.2 Website Structure and Navigation

The website is organized into a small number of main pages to keep the user experience consistent and simple on both desktop and mobile devices. Since the goal of the site is to enable users to find transportation information quickly, the navigation structure is clear and consistent across all pages.

The website pages are:

Page	Purpose
Home	Introduces RouteWise Guyana and gives quick access to the route search feature
Map	Displays all routes on a single interactive map, browsable by corridor.
Routes	List available routes for users who prefer to browse the website.
Guide	Provides basic guidance for users on route information, fares, landmarks, and how to use the system.
Feedback/ Report an issue	Allows the users to report incorrect, unclear, incomplete, or potentially inaccurate fare or route information shown on the site.
Route details	Displays detailed information for a selected route, including fare, travel time, stops, landmarks, and map information.

4.3 Data Used by the system

The system uses structured route data to display minibus information consistently across the website. This data supports route search, route directory, route details, and interactive map.

Data Field	Description	Example
Route Number/ Name	The name or description of the route.	Route 44: Stabroek Market to Mahaica
Corridors	The main travel corridor.	East Coast Demerara
Start Point and Destination	The main beginning and ending points of the route.	Stabroek Market Bus Park -> Mahaica
Landmarks and Popular places	Important stops or recognizable places along the route.	Stabroek, Kitty and Better Hope
Estimated fare and travel time	Approximated cost and duration of the journey.	GYD $100 - $260, around 1 hour
Map location data	Points used to show stops and route paths on the map.	Approximate points for Stabroek, Kitty, Better Hope and Mahaica
Notes	Extra details or including whether the data is estimated or unverified.	Approximated fare based on the 2018 data.

Each route is stored as a structured record in a JavaScript data file. Beyond the fields listed above, each route also includes a list of stops with their coordinates, the fare charged from Stabroek Market to each individual stop and a flag indicating whether the stop is a terminal or a regular stop. Some routes also include an alternate path for cases where buses on the same route do not all follow the exact same streets. This structure makes it possible to display detailed information on the route detail page while keeping the list of corridors and the general route directory simple.

5. Implementation

5.1 Client-Side Implementation

The client-side of RouteWise Guyana was implemented using three technologies: HTML, CSS and JavaScript.

The client-side implementation includes:

1. **HTML page structure** for the main website pages, including Home, Routes, Map, Route Details, Guide, and Feedback.
2. **CSS styling and layout** was used to create the overall visual design for the site as well as page spacing, route cards, form design, map sections, buttons, and responsive layout.
3. **A responsive design** that allows the same basic function of the website to exist regardless of whether it is accessed from a desktop computer or mobile device.
4. **Route filtering and sorting**, allowing users to perform a route search at either the Home page or Routes Page through JavaScript.
5. **Dynamic route cards**, where route information is loaded from the JavaScript route data file instead of being manually repeated on every page.
6. **Interactive map display** using Leaflet.js and OpenStreetMap tiles to show route paths and stop locations.
7. **Road path drawing** using the OSRM routing service so the map shows the actual road that the minibus follows, rather than only a straight line between stops. If OSRM is not available, the system falls back to drawing straight lines so that the map still works.
8. **Alternate path support** for routes that do not all follow the same streets. For example, Route 40 is drawn with a blue line for the alternate path through Thomas Street, in addition to the main red line.
9. **Client-side form validation** using JavaScript, which checks that the feedback description is long enough and that any email entered looks like a valid email before the form is sent to the server.
10. **Local storage** to save favorite routes in the user's browser. Saved routes appear on the home page so the user can return to them quickly, and clicking a saved route again removes it from favorites.

5.2 Server-side Implementation

The server-side part of RouteWise Guyana was implemented using PHP. This was mainly used for the feedback/report issue feature which allows users to send reports about incorrect fares, missing stops, map issues or general feedback.

The server-side implementation includes:

1. **PHP form processing,** where feedback submitted by users is sent to process_feedback.php for handling. The script only runs if the request was a POST, otherwise the user is redirected back to the feedback page.
2. **POST method handling**, the form data is sent to the server using the POST method and accessed through the $_POST array.
3. **Input validation**, required fields are checked before the report is accepted.
4. **Input cleaning**, using PHP functions such as trim(), strip_tags(), and htmlspecialchars() to remove unsafe or unwanted input before it is saved.
5. **Sessions** are used to prevent the same user from submitting the form too many times in a short period. After each submission, the time is stored in the session and any further submission within 30 seconds is blocked with a clear error message.
6. **Cookies** are used to remember the reporter's name for 30 days. The next time the user opens the feedback page, the name field is automatically filled in, which makes the form faster to complete for returning users.
7. **Text file storage,** reports submitted by users are appended to a file called submissions.txt.
8. **Status messages after submission**, after the PHP script handles the form, the user is redirected back to the feedback page with a value in the URL such as ?submitted=1 or ?error=missing_fields. JavaScript on the page reads this value and shows the matching success or error message to the user.

5.3 Tools and Technologies

The following tools and technologies were used in the development of RouteWise Guyana.

Tools and Technologies	Purpose in the Project
HTML	Used to structure the webpages and content.
CSS	Used to design the layout of the site, control responsive layouts across all desktop and mobile devices screens.
JavaScript	Used for interactivity, such as searching for routes, filtering routes by a selected criterion, sorting routes, displaying route information, interaction with map, form validation and favorite routes.
PHP	Used to handle the backend server-side processing of the submitted feedback form.
Leaflet.js	Used to render an interactive map that can display route markers and paths.
OpenStreetMap	The map tile provider used in Leaflet Map.
OSRM Routing Services	Used to assist with drawing route paths on the map where possible
Visual Studio Code	Used as the main code editor during development.
PHP Local server/ Localhost	Used to test the PHP feedback form processing locally.
AI-Assisted Tools	Used as support for debugging, wording, code checking, and identifying possible errors during development.

6. Usability Evaluation

6.1 Purpose of Evaluation

The purpose of the usability evaluation was to assess whether RouteWise Guyana allows users to find route and fare information easily, quickly, and with minimal confusion.

The evaluation focused on:

- Whether users can complete their intended task with ease.
- Whether users can locate the desired route and fare information using an acceptable number of steps.
- Whether RouteWise Guyana functions properly on both desktop and mobile devices.
- Whether users are able to interpret the route details, fare information, landmarks, and map displays correctly.
- Whether users can clearly complete the feedback/report issue form provided.
- Whether RouteWise Guyana meets the needs of commuters, students, tourists, and first time users.

This is significant as a website could function correctly from a technical standpoint yet continue to provide poor user experiences if users find it difficult to use. This evaluation considers whether the system is effective, efficient, and satisfactory for its intended users.

6.2 Participants

The usability evaluation involved a small group of participants who represent the likely users of RouteWise Guyana.

The participant group included:

- University students who use minibuses to travel to and from campus.
- Commuters that regularly take the same routes as part of their daily commute.
- Less experienced users who have had little experience of minibus services.
- First time users/visitors that are reliant on landmarks and guidance.
- At least one user accessing the website on a mobile device.
- At least one user that has access to a Desktop/Laptop Computer.

6.3 Test Tasks

Participants were asked to complete a series of real-world tasks using RouteWise Guyana. These tasks were based on the main features of the website, and represent the kinds of information users would typically need when either planning a trip or checking a minibus route.

The test tasks were:

1. Find a route using the website

The participant searched for a route on the website by entering or selecting a destination or route number.

2. View route details

The participant opened one of the selected routes and identified the route number, destination, estimated fare, and travel time.

3. Identify major stops and landmarks

The participant used the route details page to identify at least two major stops or landmark locations along the selected route.

4. Use the route directory

The participant opened the Routes page and used the available search, filter or sort options to locate a particular route or corridor.

5. Use the interactive map

The participant opened the Map page and viewed one of their selected routes displayed in the map display area.

6. Submit a feedback/report issue form

The participant completed the Feedback form using sample information to report an incorrect fare, missing stop, unclear route details, or other issue.

The tasks above were selected as they represent the main actions users are expected to perform on the website. These tasks enabled the group to assess if users could easily access the required information, understand the details of each route, effectively utilizing the map display, and successfully complete the feedback process without major difficulty.

6.4 Metrics Used

The usability evaluation used simple metrics to measure how well users can complete the main tasks on RouteWise Guyana. These metrics helped the group assess whether the website is effective, efficient, and satisfactory for its intended users.

Metric	Purpose	Calculation/Method
Task Success Rate	Measures whether users completed each task successfully.	$(Number\ of\ successful\ completions\ \div\ Total\ task\ attempts) \times 100$
Average Time on Task	Measures how long users took to complete each task.	$Total\ time\ taken \div Number\ of\ participants$
Average Clicks/Taps	Measures how many steps users needed to find a route or fare information.	$Total\ clicks\ or\ taps\ \div\ Number\ of\ participants$
Error Rate	Records mistakes such as selection of incorrect routes, clicking of the wrong page, or the misinterpretation of route information.	$(Total\ number\ of\ errors\ \div\ Total\ task\ attempts) \times 100$
User Satisfaction Rating	Measures how easy or difficult users felt the website was to use.	*Average rating using a 1 - 5 scale*

The following scale was used for the satisfaction rating:

Rating	Meaning
1	Very Difficult
2	Difficult
3	Neutral
4	Easy
5	Very Easy

User comments and observations were also recorded to explain any issues behind the numbers, such as confusing labels, missing information, layout problems, or features users found helpful.

6.5 Findings

The usability evaluation results were organized using task success rate, average time on task, error rate, and user satisfaction rating. The assessment included six participants who executed six primary tasks on RouteWise Guyana. The full usability dataset, including participant profiles and detailed task data, is provided in the appendix.

6.5.1 Task Success Results

Task success rate was used to assess whether participants were able to complete each task successfully. This helped the group assess the effectiveness of the website.

Table 1: Task Success Results

Task	Description	Successful Users	Total Users	Success Rate
Task 1	Find a route using the website	6	6	100.0%
Task 2	View route details	6	6	100.0%
Task 3	Identify major stops and landmarks	5	6	83.3%
Task 4	Use the route directory	6	6	100.0%
Task 5	Use the interactive map	4	6	66.7%
Task 6	Submit a feedback/report issue form	5	6	83.3%

The task success rate was calculated using the formula:

$$Task\ Success\ Rate\ = \left(\frac{Number\ of\ Successful\ Completions}{Total\ Task\ Attempts}\right) \times 100$$

The results show that users were most successful when finding a route, viewing route details, and using the route directory, all three of these tasks achieved a success rate of 100%. In contrast, the least amount of success was recorded for Task 5, where only 4 out of 6 users successfully used the interactive map. This suggests that the map feature may require clearer instructions, more visible labels, or simpler interaction controls.

6.5.2 Average Time on Task

The average time on task was used to help measure how long users took to complete each task. This helped the group to assess the efficiency of the website.

Table 2: Average Time on Task		
Task	**Total Time/s**	**Average Time/s**
Task 1	255	42.5
Task 2	194	32.3
Task 3	326	54.3
Task 4	261	43.5
Task 5	435	72.5
Task 6	333	55.5

The average time on task was calculated using the formula:

$$Average\ Time\ on\ Task = \frac{Total\ Time\ Taken}{Number\ of\ Participants}$$

Task 5, which involved using the interactive map, had the longest average completion time at 72.5 seconds. This supports the results from the task success rate, as the map task also had the lowest success rate. This suggests that some participants required additional time to learn how to interact with or interpret the routes displayed on the interactive map.

On the other hand, task 2 which involved route details, had the shortest average completion time. This suggests that accessing the route details information provided by the website was easier for participants to access and understand.

6.5.3 Error Rate

The error rate was used to record errors such as selection of incorrect route, clicking of the wrong page, or the misinterpretation of route information.

The overall error rate was calculated using the formula:

$$Error\ Rate\ = \left(\frac{Total\ Number\ of\ Errors}{Total\ Task\ Attempts}\right) \times 100$$

Total number of errors recorded:

$$2 + 1 + 4 + 1 + 0 + 3 = 11$$

Since each of the six participants completed six tasks each, the total number of tasks was:

$$6 \times 6 = 36$$

Therefore, the overall error rate was:

$$\left(\frac{11}{36}\right) \times 100 = 30.6\%$$

Although the users successfully completed the majority of the tasks, errors still occurred along the way. It should be noted that this figure represents the total number of recorded errors as a proportion of all task attempts, not the percentage of tasks that failed. A single task attempt can contain more than one error, which is why the error rate appears high alongside a successful completion rate of 88.9%. Most of the errors appear to be related to the interactive map, landmark identification, and the feedback or report issue form. These areas may need design improvements to reduce confusion.

6.5.4 User Satisfaction Rating

After completing the tasks, participants evaluated the usability of RouteWise Guyana on a scale from 1 to 5.

The satisfaction scores were:

$$4 + 4 + 3 + 4 + 5 + 3 = 23$$

The average satisfaction rating was calculated using the formula:

$$Average\ Satisfaction = \frac{Total\ Satisfaction\ Score}{Number\ of\ Participants}$$

$$\frac{23}{6} = 3.8\ out\ of\ 5$$

An average satisfaction rating of 3.8 out of 5 suggests that users generally found RouteWise Guyana useful and fairly easy to use. However, the scores also show there is clear potential for improvement, especially for users who are less familiar with the minibus system.

6.5.5 Overall Summary of Results

The overall usability results are summarized below:

Table 5: Summary

Metric	Result
Overall Task Success Rate	88.9%
Overall Error Rate	30.6%
Average Satisfaction Rating	3.8
Task with Highest Success Rate	Task 1, Task 2, Task 4
Task with Lowest Success Rate	Task 5: Use the interactive map
Task with Longest Average Time	Task 5: 72.5 Seconds

The overall task success rate was determined as follows:

$$Total\ Successful\ Completion = 6 + 6 + 5 + 6 + 4 + 5 = 32$$

$$Total\ Task\ Attempts = 6\ \times 6 = 36$$

$$\left(\frac{32}{36}\right) \times 100 = 88.9\%$$

The overall success rate of 88.9% suggests that RouteWise Guyana was generally effective in helping users complete the main tasks. Most users were able to search for routes, see route details, and use the route directory with little difficulty.

However, the interactive map had the lowest success rate and the longest average completion time. This suggests that the map feature may need clearer labels, better instructions, or a simpler design.

The majority of users who attempted to complete the Feedback/Report Issue Form were successful; however, one user failed to do so. This may indicate that there is room for improvement in making the instructions provided for completing the form clearer and providing better confirmation messages.

The findings show that RouteWise Guyana meets the main usability goals and requirements by allowing users to search for routes, view route details, and access fare information without much difficulty.

The findings also relate to general usability principles such as those outlined in Nielsen's heuristics. For example, the difficulty some participants experienced with the interactive map relates to visibility and user guidance, since users appear to need more support when interpreting route paths and map markers. The reliance on landmarks also shows the importance of recognition rather than recall, as users are more likely to understand familiar places than exact route names or street names. The feedback form and route details pages performed better because they presented information in a more direct and structured way.

Overall, the results indicate that RouteWise Guyana is functional and generally easy to navigate. The main areas for improvement are the map interface, landmark identification, feedback form guidance, and mobile layout, as these would help reduce errors and better support first time users.

7. Data Source and Use of External Tools

Data Source

The route and fare information used in RouteWise Guyana was based on selected public transportation routes in Guyana. The data provided by RouteWise Guyana includes route numbers, destinations, corridors, major stops, landmarks, estimated fare, estimated travel times, and map location points.

The route information was determined from publicly listed routes. Group members also drew on their personal travel experiences, since many local minibus routes are understood through daily usage, knowledge of common areas such as bus parks, landmarks, and general commuter knowledge rather than through one complete official digital route database.

Fare information was estimated using the 2018 Ministry of Business/ United Minibus Union fare structure where available. This fare structure applied to several minibus zone/routes and became effective on September 1, 2018 (Department of Public Information, 2018). Recent reports also indicate ongoing fare-related disputes, with some operators requesting fare increases, after claims that fares have not increased since 2018 (News Room Guyana, 2026).

Because some of the route's fare and travel time information may not be exact, RouteWise Guyana presents the information as guidance rather than a definitive official transport platform. Users are also advised to verify their final fare with the conductor prior to traveling.

Use of External Tools

Tools/ Resources	Use in the Project
Custom Images and Design Assets	Used to enhance the presentation of the website.
Leaflet.js	Used to render an interactive map that can display route markers and paths.
OpenStreetMap	The map tile provider used in leaflet Map.
OSRM Routing Services	Used to assist with drawing route paths on the map where possible
Visual Studio Code	Used as the main code editor during development.
PHP Local server/ Localhost	Used to test the PHP feedback form processing locally.

8. Limitations and Future Improvement

8.1 Limitations

1. This system is designed for educational use and is not intended to represent a fully official public transportation platform.
2. Due to limited project timeline, the first version of the website focuses on selected routes and major commuter corridors rather than every minibus route in Guyana.
3. Some route, fare and travel-time information may be approximated because digital public transportation data is limited or not easily available.
4. The site does not provide live GPS tracking of minibuses.
5. The site does not show real-time bus arrival or departure updates.
6. The system does not include payment options or account logins.
7. The accuracy of the information depends on the quality of the observation, commuter input and publicly available sources used by the group.

8.2 Future Improvement

1. More verified minibus routes could be added to cover a wider area of Guyana.
2. Routes and fare information could be updated though partnerships with transport authorities, Minibuses Associations, or other official sources.
3. The map feature could be improved with more accurate route paths, stops and landmarks.
4. A live update feature could be added to show delays, route changes, or fare changes.
5. The system could offer offline viewing capabilities so users can view saved route information without constant internet access.
6. Feedback submitted from users could be evaluated and considered in updating the route database to increase the accuracy of the route database.
7. A future version could include multilingual support or simpler language options for tourists and first-time users.

9. Conclusion

RouteWise Guyana was developed in response to the lack of clear and easily available digital information on minibus routes, fares, stops, and landmarks in Guyana. The project demonstrates how HTML, CSS, JavaScript, and PHP can be used to develop a web-based application, which is both responsive and provides features such as route searching, route filtering, detailed route information, a dynamic map, ability to save favorite routes, and to submit a feedback/report issue form.

Results from the usability evaluation demonstrated that the system successfully supported users in completing most of their required tasks such as finding a route, displaying route information and navigating through the route directory. Additionally, the results also suggested that areas such as the interactive map feature, landmark directions and submitting user feedback have potential for improvement to assist first-time users.

RouteWise Guyana is a functional prototype for collecting and disseminating public transportation data specific to a local Guyanese context and allows opportunities for future development based on more verified routes, accurate fares, real time update capabilities, or further partnerships with transportation authorities and minibus associations.

10. References

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11. Appendices

Appendix A: Participant Profile

Table 1: Participant Profile			
Participant	**User Type**	**Device Used**	**Experience with Minibus Routes**
P1	University Student	Mobile phone	Moderate
P2	Regular Commuter	Mobile phone	High
P3	First-time user/visitor	Mobile phone	Low
P4	University Student	Laptop	Low
P5	Regular Commuter	Laptop	High
P6	Less experienced user	Laptop	Low

Appendix B: Detailed Usability Evaluation

Table 2: Detailed Task Data									
Participant	**Devices**	**Task 1 Time**	**Task 2 Time**	**Task 3 Time**	**Task 4 Time**	**Task 5 Time**	**Task 6 Time**	**Total Errors**	**Satisfaction/5**
P1	Mobile phone	42	31	54	47	75	58	2	4
P2	Mobile phone	34	28	41	39	63	46	1	4
P3	Mobile phone	61	45	78	59	102	72	4	3
P4	Laptop	36	27	46	34	55	49	1	4
P5	Laptop	29	24	38	31	49	43	0	5
P6	Laptop	53	39	69	51	91	65	3	3

Appendix C: Usability Calculations

Table 3: Task Success Results				
Task	**Description**	**Successful Users**	**Total Users**	**Success Rate**
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Metric	**Result**
Overall Task Success Rate	88.9%
Overall Error Rate	30.6%
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Task with Lowest Success Rate	Task 5: Use the interactive map
Task with Longest Average Time	Task 5: 72.5 Seconds

Appendix D: Wireframe/ Interface Design

The Figma design can be accessed here:

https://www.figma.com/design/Cwj42ZGDHTyQJ2E10afBNQ/RouteWise?node-id=0-1&t=csjNyfH5NZh2SIeH-1

Appendix E: Lecturer Approval

Project Approval Request – CSE2201 Group Project

Convert to PDF

Ravindra Persaud Tue, Apr 28, 10:42 AM

Dear Sir, I hope you are doing well. I am writing on behalf of our group to seek approval for our project idea for CSE2201. The problem we have identified is th

Lenandlar Singh Tue, Apr 28, 11:45 AM

to me

Hi Ravindra

This is interesting and potentially very useful. Where do you plan to get your info from?

Go ahead

Len

Appendix F: Feedback Submission/ Form Evidence

submissions.txt

```
# RouteWise Guyana - Feedback Submissions Log
# Entries are appended by process_feedback.php
========================================
Submitted:   2026-05-23 18:03:36
Name:        Aaliyah Singh
Email:       aaliyah.singh24@gmail.com
Category:    Fare Overcharging
Route:       Route 44 (Mahaica via UG)
Date/Time:   2026-05-19T14:03
Plate:
Description: I was charged more than the fare shown for Route 44.

========================================
Submitted:   2026-05-23 18:04:58
Name:        Daniel Persaud
Email:       danielpersaud17@gmail.com
Category:    Missing Stop/Landmark
Route:       Route 40 (Kitty / Campbellville)
Date/Time:   2026-05-19T15:04
Plate:
Description: Kitty Market should be added as a clearer landmark.

========================================
Submitted:   2026-05-23 18:07:37
Name:        Renuka Singh
Email:       renukasingh21@gmail.com
Category:    Map Glitch or Bug
Route:       Route 42 (Timehri / CJIA)
Date/Time:   2026-05-20T16:06
Plate:
Description: I was not sure where to click first on the map and there were two routes.

========================================
Submitted:   2026-05-23 18:08:33
Name:        Alicia Fredericks
Email:       aliciafredericks@gmail.com
Category:    General Feedback
Route:       Route 48 (Sophia)
Date/Time:   2026-05-20T16:30
Plate:
Description: The required fields should be easier to identify.
```